



Chandrayaan 2 or UFO?

Chandrayaan 2 was spotted by Australian civilians and they thought it was a UFO! <https://digit.in/aug19-21>



Recreating Marsquakes

Seismologists conducted an experiment that allowed them to experience what a Marsquake feels like. <https://digit.in/aug19-22>

Three repeating FRB sources are known as of today.

After their initial discovery, FRBs were added as objects of interest to the Breakthrough Listen initiative, and the project has detected fifteen repeated bursts from the source known as FRB 121102. The observations were collected from the Green Bank Telescope in West Virginia, which collected 400 TB of data in just five hours of observations. Using the Parkes Radio Telescope, Listen also identified an entirely new FRB source. The observations from the Listen project is expected to help scientists better understand the mysterious phenomena, even if the results are not aligned with the goals of the project. In previous instances, an unexpected, repeating signal was interpreted as a sign of extraterrestrial intelligence.

The first pulsar ever identified, PSR B1919+21 was also designated as LGM 1 for "Little Green Men". The researchers were initially mystified by the radio pulses from the source repeating with surprisingly regular periodicity. The second such source, KIC 8462852 was also designated as LGM 2, and showed extreme periodic dimming, leading some scientists to believe that it could be surrounded by a Dyson structure to extract energy from the Star. The explanation was something much more mundane than an alien megastructure - dust. Eventually, there will be a source discovered where all natural processes can be ruled out.

EXOPLANETS

Listen works by using its considerable \$1 billion funding to buy telescope time from observatories around the world. The instruments are then trained to the targets selected by the initiative. The collected data is then made available to scientists, the signal processing community as well as those users who still run SETI@home through the distributed computing project hosted by Berkeley. Breakthrough Listen is just one of the Breakthrough Initiatives, and they are

actively searching for habitable exoplanets as well. Breakthrough Starshot is another ambitious project to send interstellar probes. Towards this end, Breakthrough has purchased time from the Very Large Telescope (VLT), to specifically look for planets in the habitable zone in a single system - Alpha Centauri, the closest star system to the Sun, at a distance of 4.37 light years. Nanocraft can make it to this system within a single generation. The system has three stars, and the VLT is equipped to find any rocky planets in the habitable zones of any of these stars. In 2016, a planet in the habitable zone of Alpha Centauri C, or Proxima Centauri was found. However, subsequent studies indicated that the planet was inhospitable for life as we know it on Earth, because of constant ravaging from the host star. The Automated Planet Finder, which will study 10 stars a day for signs of exoplanets, over a period of 10 years, has also collaborated with the Breakthrough Listen project to simultaneously search these same targets for light or laser signals.

TECHNOSIGNATURES

Technosignatures that can be detected over interstellar distances include communication beacons, rocket trails or Clarke exobelts - rings of satellites around planets. By contrast, finding alien life forms through bio signatures involves probing the atmospheres of exoplanets for signs of methane, or disequilibrium of the gases in the atmosphere, that cannot be explained by natural sources. Breakthrough Listen has announced a collaboration with the VERITAS (Very Energetic Radiation Imaging Telescope Array System) array of four 12-meter telescopes, located at the Fred Lawrence Whipple Observatory in Arizona.

"Breakthrough Listen is already the most powerful, comprehensive, and intensive search yet undertaken for signs of intelligent life beyond Earth. Now, with the addition of VERITAS, we're sensitive to an important new class of signals: fast optical

pulses. Optical communication has already been used by NASA to transmit high definition images to Earth from the Moon, so there's reason to believe that an advanced civilization might use a scaled-up version of this technology for interstellar communication," said Dr Andrew Siemion, who leads the Breakthrough Listen side of the collaboration from the University of California, Berkeley.

The VERITAS telescope is one of the most powerful telescopes in the world for studying high energy astrophysics, and is typically used for observing gamma rays. However, turns out that the telescope array is suitable for the goals of the Breakthrough Initiative, even though the instrument was originally not designed for that purpose. The array will be used for detecting a very specific type of technosignature - laser pulses, or beacons that are of a short duration, a few nanoseconds. The VERITAS instrument has previously been used to scan for such short optical pulses from KIC 8462852, as well.

Lasers can be used in space for a number of reasons, including transmitting data, controlling robotic probes, broadcasting a timestamp or to locate an object. These laser flashes, even if they last a billionth of a second, would outshine any nearby stars, provided they are aimed straight at the Earth. The VERITAS team has narrowed down the search to 120 target stars in particular. Each star will be observed for a duration of about fifteen minutes, and if any laser pulses are detected, the observation time will be extended to rule out false positives. The important factor here is the confirmation from all four telescopes in the array.

Yuri Milner, founder of Breakthrough Initiatives said, "When it comes to intelligent life beyond Earth, we don't know where it exists or how it communicates. So our philosophy is to look in as many places, and in as many ways, as we can. VERITAS expands our range of observation even further." 